

HSIU-CHI LIU

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Summary

Go-focused software engineer building reliable distributed systems and cloud-native platforms. Experienced in microservice orchestration, gRPC service modernization, Kubernetes and AWS infrastructure, and production observability with OpenTelemetry. Interested in cloud infrastructure, runtime systems, SRE, and AI platform engineering.

Experience

Software Engineer, Trend Micro, Taipei

Oct. 2025 - Present

- Improved a distributed microservice orchestration system, raising end-to-end workflow success rate to **99%** through reliability-focused service and workflow changes
- Built production observability with **OpenTelemetry** and the **LGTM** stack, enabling end-to-end tracing and standardized metrics across distributed service workflows
- Designed and migrated a legacy **C++** gRPC server to a **Go** gRPC service using **cgo**, modernizing the runtime while preserving interoperability
- Built cloud-native deployment and operations workflows with **AWS**, **Terraform**, **Kubernetes**, **Helm**, **Prometheus**, and **Grafana**

Software Engineer Intern, Hewlett-Packard, Taipei

Jul. 2024 - Jul. 2025

- Built a RESTful data integration API using Go and Gin, enabling internal systems to exchange departmental data through a unified backend interface
- Improved internal knowledge retrieval by developing a Retrieval-Augmented Generation (RAG) database for faster access to department-specific information

Projects

Cometline (Personal Side Project)

- Built a personal AI platform spanning an **Electron** desktop shell and a **Go** agent runtime, abstracting multiple LLM providers behind a unified interface for streaming responses, tool execution, web search, and semantic memory
- Designed multi-step agent orchestration with workspace-isolated sessions, workspace-scoped file and shell tools, and computer-use automation
- Implemented runtime components for execution lifecycle management and streaming event delivery

go-symphony (Personal Side Project)

- Reduced Go web service setup effort by building an LLM-friendly CLI with Cobra and Bubble Tea for opinionated Gin, PostgreSQL, Supabase, and sqlc scaffolding
- Accelerated full-stack project bootstrapping by adding out-of-the-box Supabase initialization and optional SvelteKit or Next.js frontend scaffolds

Personal Blog with Comment System (Personal Side Project)

- Built a full-stack blog comment system using Go, Gin, sqlc, and PostgreSQL to support persistent user discussions
- Enabled real-time user interaction by implementing WebSocket-based comment updates and OAuth-secured authentication

Open Source Contribution

KubeRay

Open-source Kubernetes operator designed to simplify the deployment and management of Ray applications.

- PR #4038:** Improved test maintainability by consolidating duplicated support utilities across KubeRay test files
- PR #4044:** Modernized Go test code by removing redundant range variable captures using Go 1.22 scoped iteration semantics

Temporal

Durable Execution platform that abstracts away the complexity of building distributed systems.

- PR #813:** Fixed Helm chart testing and added support for test pod annotations

Education

National Taiwan University, M.S. in Computer Science

Sep. 2023 - Jul. 2025

- Relevant Knowledge:** Operating Systems Design and Implementation, Computer Vision with Deep Learning, Web Development

Skills

- **Programming Languages:** Golang (primary); Python, C/C++, TypeScript
- **Distributed Systems & Backend:** gRPC, Gin, REST, WebSocket, PostgreSQL, SQL
- **Cloud & Infrastructure:** AWS, Kubernetes, Helm, Terraform, Docker
- **Observability & Reliability:** OpenTelemetry, Prometheus, Grafana, LGTM stack
- **AI Systems:** LLM orchestration, RAG, semantic memory